UEFA EURO Soccer Analysis Report

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Abstract

This project analyzes historical data from the UEFA European Championship from 1960 to 2024. We cleaned and combined multiple datasets, including match summaries, player lineups, coaching information, qualifying matches, and tournament statistics. Our analysis examines long-term trends in scoring, discipline, and attendance; compares physical attributes across player positions; studies the effect of coaching continuity; and tests whether qualifying performance can predict EURO outcomes using a Random Forest model. The results show clear historical trends, modest differences in player attributes, weak relationships between coaching stability and performance, and limited predictive power from qualifying data. This work highlights both the insights and challenges of long-term sports analytics.

Introduction

In this project, we examined how soccer has changed in the UEFA European Championship from 1960 to 2024. We worked with a large dataset that included tournament summaries, team lineups, coaching information, qualifying matches, friendly matches and detailed match files for each EURO year. The aim of this project was to understand long-term trends in scoring and discipline, explore physical differences between player positions, evaluate the effect of coaching continuity and build a simple model to test whether qualifying performance can predict tournament outcomes.

Before we could begin any analysis, we spent time preparing and cleaning the data because it contained missing values, inconsistent formats and different structures across years. After cleaning, we completed each task and created visualizations to support our findings. This

report explains the methods we used, the results we obtained, and the constraints we faced along the way.

Methodology

We started by loading all CSV files from the dataset. The files were distributed across folders such as EURO match files for different years, Nations League data, qualifying matches, friendly matches and general tournament summary files. We combined all EURO match files into a single dataset and created team-level summaries for each tournament. This allowed us to compute goals scored, goals conceded, wins, losses, draws, goal difference and points for every team in every EURO year.

The dataset required extensive cleaning. Many values were missing, especially height, weight, player birth dates, stadium conditions and some coaching fields. Instead of dropping any rows, we created "was_missing" flags for every column that had missing values. This allowed us to preserve all historical data while keeping track of missingness. We cleaned several data types, converted dates into proper `datetime` format, normalized team codes to uppercase and transformed numeric strings into integers or floats. We also standardized player positions. The dataset used many detailed labels for positions, so we mapped these into four main groups: goalkeeper, defender, midfielder and forward.

For Task 1, we used the cleaned tournament summary table to examine changes in goals, red cards and attendance. We created line plots for each metric and printed summary statistics below each plot.

Task 2 used the lineup dataset. After mapping player positions, we created box plots for height and weight across the four main positions and included summary tables.

Task 3 required merging coach data with team performance. We created two features: one to measure whether a team kept the same coach from the previous EURO, and another to measure how much EURO experience each coach had. We merged these features with team performance tables to study patterns.

Task 4 involved predictive modeling. We first computed qualifying performance metrics by aggregating goals, wins, losses and points for each team. We then merged qualifying metrics with EURO performance metrics. We trained a Random Forest model to predict different indicators of EURO performance and evaluated it using MAE and R^2. We also created feature importance charts to understand which qualifying metrics mattered most.

Results and Outcomes

Task 1: Trends in Goals, Penalties, and Attendance

The plot for goals per match showed that scoring has increased over time. The average goals per match across tournaments was about 2.63. The highest value, around 4.75, appeared in 1976, while the lowest, about 1.40, appeared in 1968. The graph suggested that modern tournaments are more attacking, possibly because of tactical evolution and changes in rules.

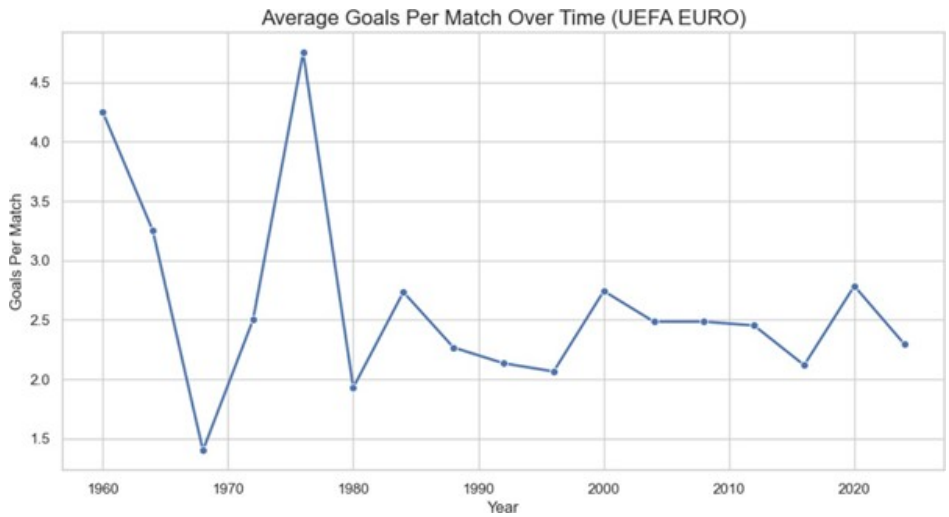

Figure 1: Average Goals Per Match Over Time (UEFA EURO).

From the graph, it is clear that scoring trends rise gradually over time. Modern tournaments show more attacking play. Some early tournaments show unstable numbers because they had fewer matches.

The chart for red cards per match showed mostly low values through all years, with small spikes in the 1990s. Recent tournaments showed fewer red cards.

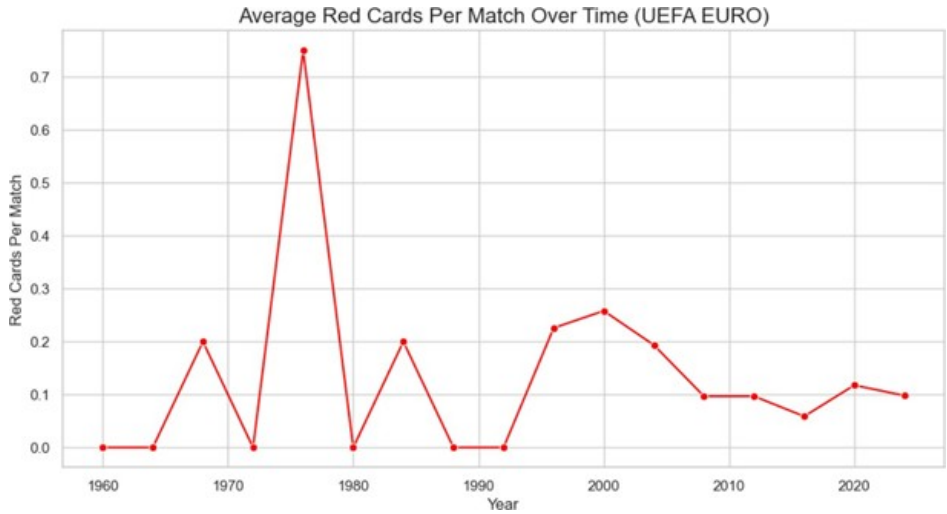

Figure 2: Average Red Cards Per Match Over Time (UEFA EURO).

The graph shows that red cards remain low in most editions. Only a few years show noticeable peaks. Overall, discipline has stayed consistent.

The attendance chart showed a clear rise, especially after the 1990s, as stadium capacities increased and the tournament gained popularity.

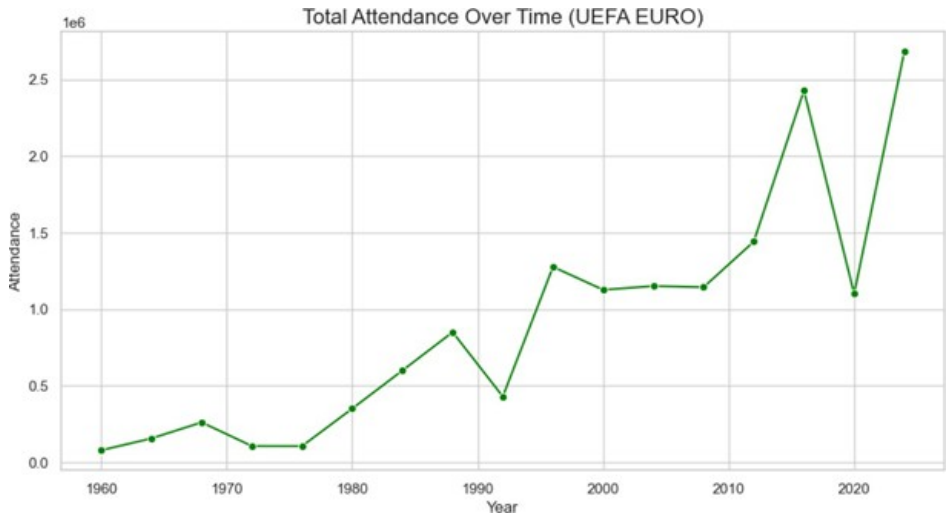

Figure 3: Total Attendance Over Time (UEFA EURO).

The graph clearly shows rising attendance, especially after the 1990s. Missing attendance values in early years create slight gaps, but the trend is still clear. One limitation in this task was missing attendance data in some early tournaments. Another issue was the small number of matches in early EUROs, which made their averages less stable.

Task 2: Player Physical Attributes by Position

The results showed that goalkeepers were the tallest and heaviest group. Defenders were also tall and often heavier than midfielders. Midfielders had more average values for height and weight, while forwards showed the widest range. Some forwards were tall target strikers, while others were shorter, faster attackers.

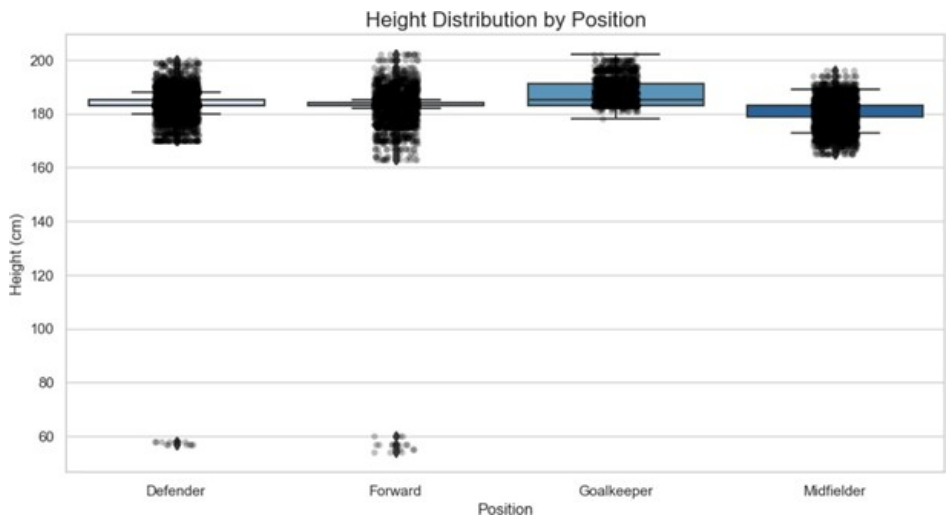

Figure 4: Height Distribution by Position.

The graph shows that goalkeepers and defenders dominate the upper height ranges. Midfielders and forwards fall in the middle, with forwards showing more variation.

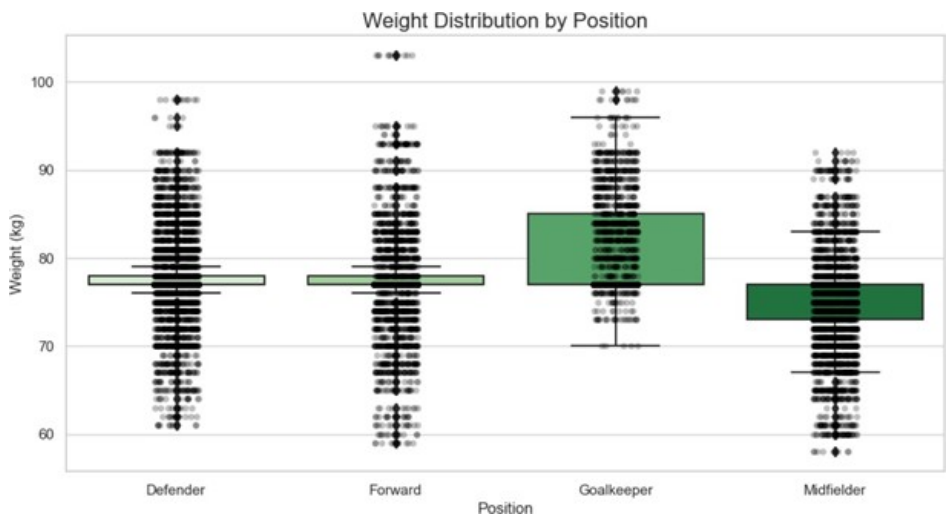

Figure 5: Weight Distribution by Position.

The weight distribution follows similar trends. Goalkeepers and defenders are heavier. Midfielders and forwards are more varied, which reflects the different roles within those

positions.

The box plots were not very sharp because many height and weight values were missing, especially before 1990. Many players had unknown positions, which made grouping harder. Birth dates were missing in many rows, so we could not calculate age properly. Because of these issues, we could only make general observations about player physical differences. The limitations prevented deeper analysis.

Task 3: Coaching Continuity and Team Performance

When we explored the coaches dataset, we found a major limitation. The dataset did not include coach nationality. Instead, the "country" field represented the team's country, not the coach's origin. This meant we could not compare local coaches with foreign coaches.

Since nationality could not be used, we focused only on coaching continuity. We created a feature that showed whether a team kept the same coach from the previous EURO. We also created a feature that counted how many EURO tournaments a coach had led before.

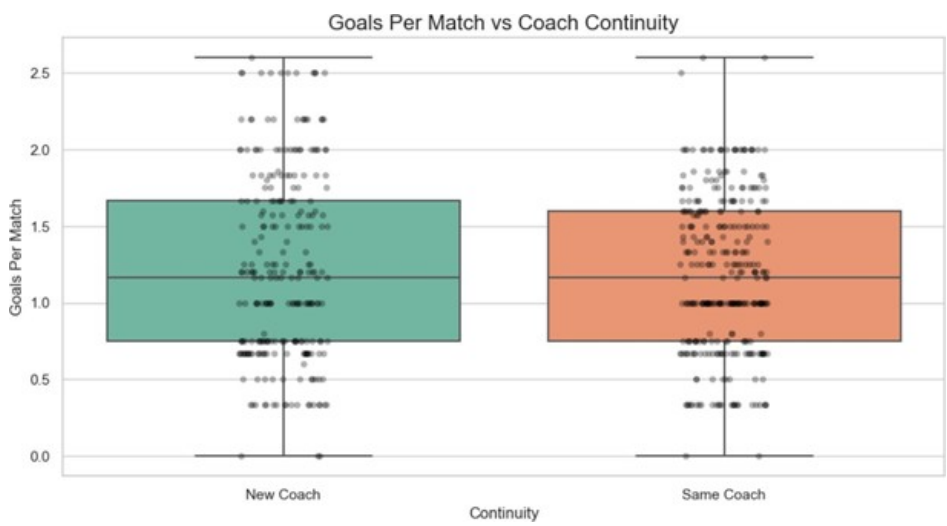

Figure 6: Goals Per Match vs. Coach Continuity.

The graph shows small differences between teams with a new coach and teams with the same coach. The effect is present but not strong.

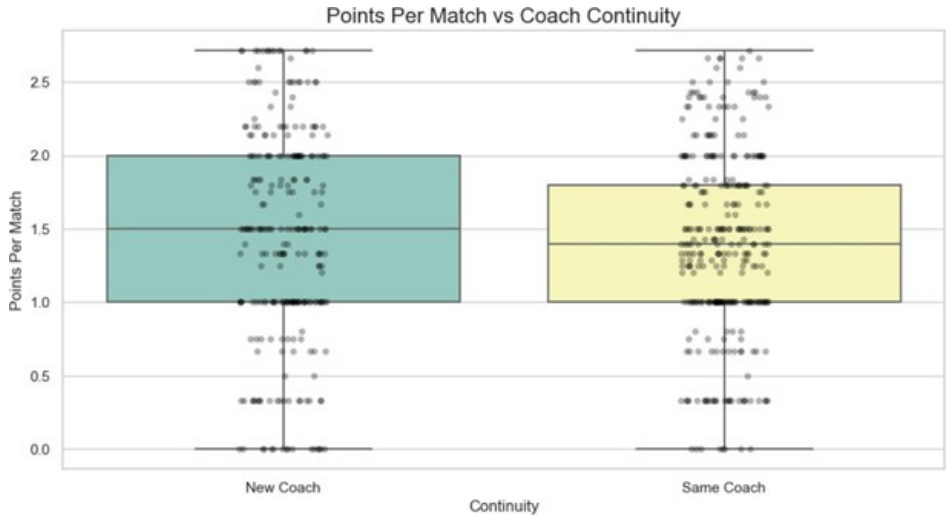

Figure 7: Points Per Match vs. Coach Continuity.

The difference in points per match is small. Some teams do slightly better with continuity, but the variation is large.

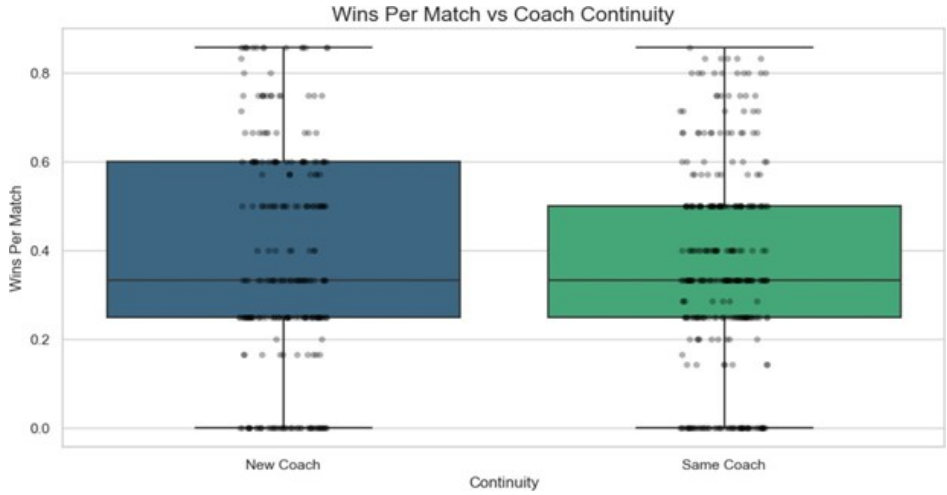

Figure 8: Wins Per Match vs. Coach Continuity.

There is no strong separation between the two groups. EURO tournaments are unpredictable, so continuity does not guarantee success.

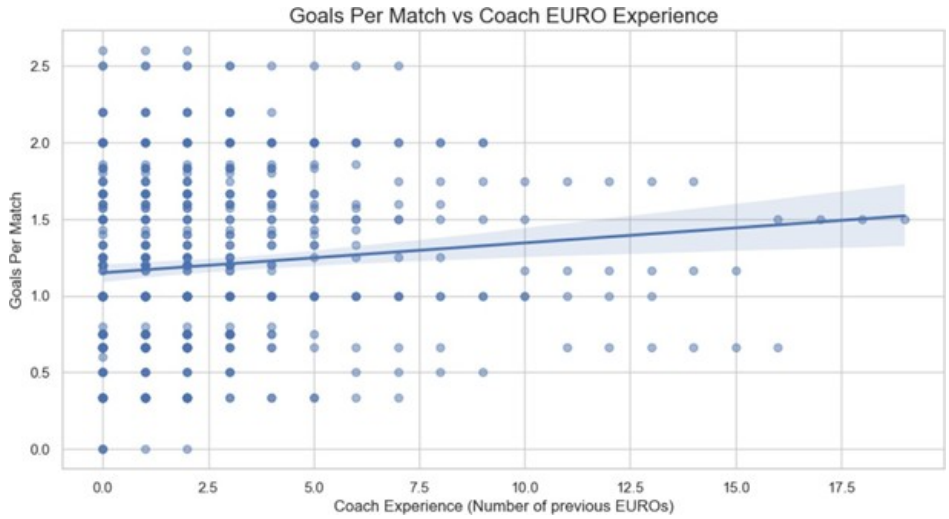

Figure 9: Goals Per Match vs. Coach EURO Experience.

The regression line shows a mild upward trend, but the scatter is wide. This means coaching experience helps a little, but not enough to predict outcomes.

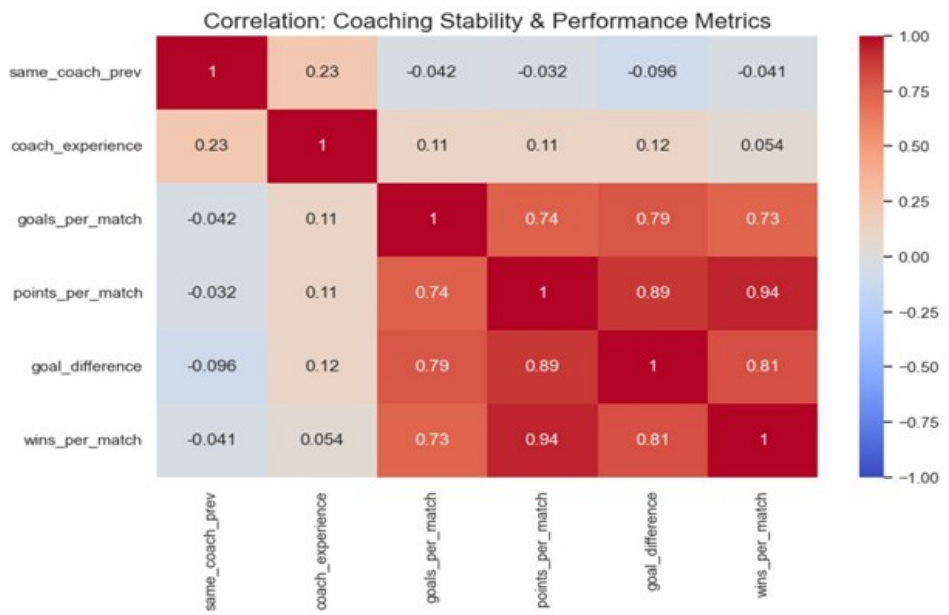

Figure 10: Correlation: Coaching Stability and Performance Metrics.

The heatmap shows very weak correlations. None of the features strongly predict performance.

After merging these features with team performance, we found that teams with the same coach sometimes had slightly better results, but the differences were small. Coach experience also showed a mild positive trend, but the relationship was weak. The main constraints were limited data, because EURO happens only every four years, and frequent coaching changes

between tournaments. Some early tournaments had incomplete coach information as well. These issues made the results less reliable.

Task 4: Predicting EURO Success from Qualifying Performance

In this task, we built a Random Forest model to predict EURO performance using qualifying performance. We trained one model for each target: EURO goals per match, EURO points per match, EURO wins per match and EURO goal difference.

The results showed that the model did not perform well. All R^2 scores were negative. For example, predicting EURO goals per match produced an R^2 of about -0.27, and predicting EURO goal difference produced an R^2 of about -0.46. A negative R^2 means the model performed worse than simply predicting the average for every team.

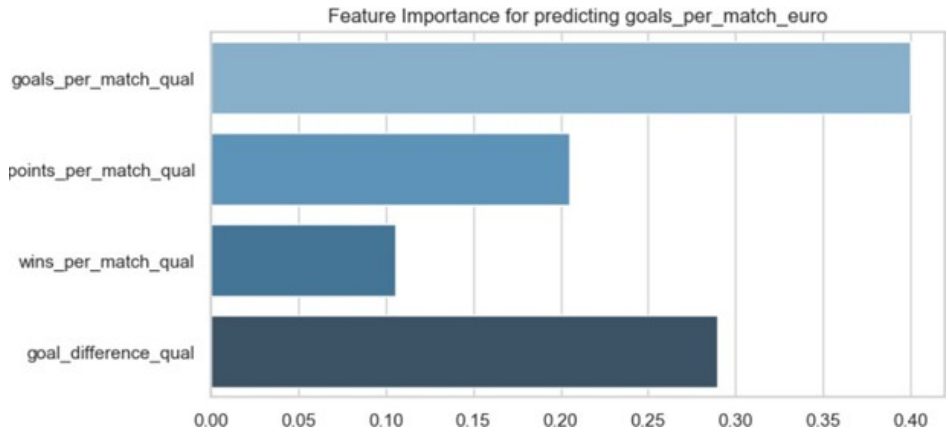

Figure 11: Feature Importance for Predicting EURO Goals Per Match.

Qualifying goal difference and qualifying goals per match are the strongest predictors, but they are still not strong enough to produce accurate predictions.

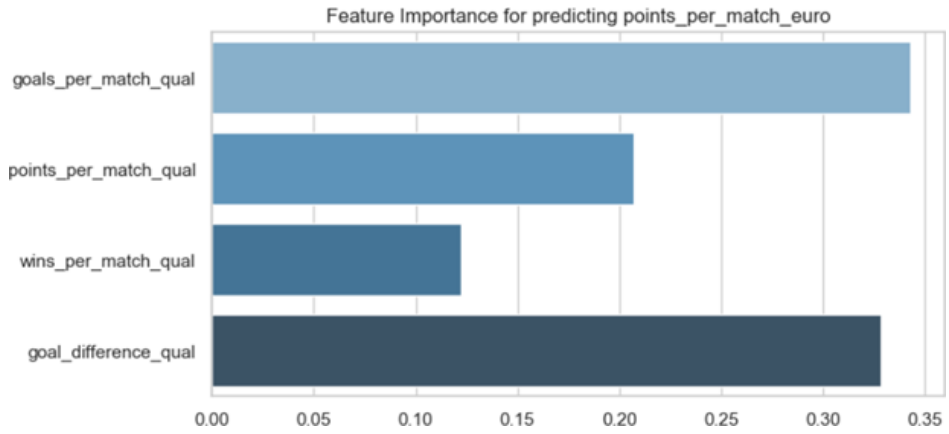

Figure 12: Feature Importance for Predicting EURO Points Per Match.

The model shows similar importance patterns. Wins per match is again the weakest feature because qualifying groups vary in difficulty.

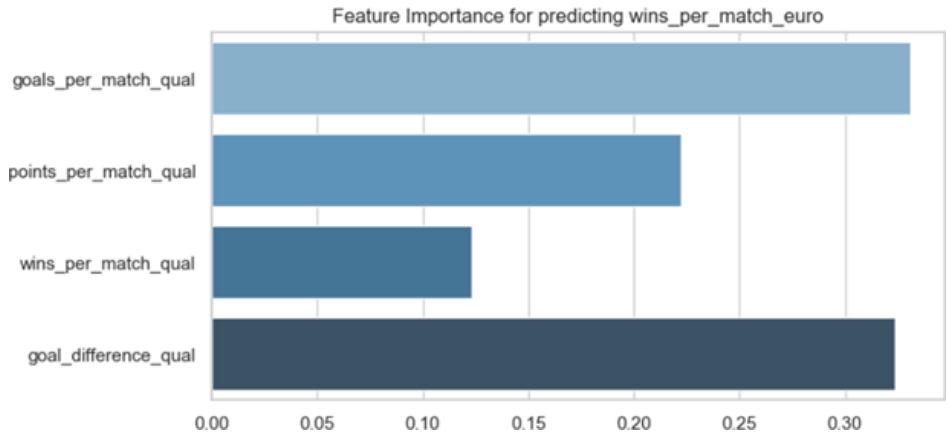

Figure 13: Feature Importance for Predicting EURO Wins Per Match.

The chart confirms that qualifying performance does not strongly predict wins in EURO tournaments.

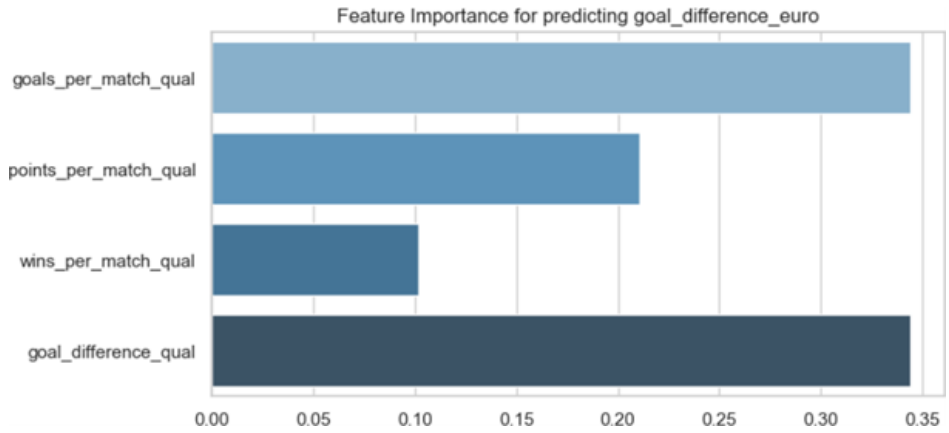

Figure 14: Feature Importance for Predicting EURO Goal Difference.

Even though goal difference is the strongest predictor, the model still fails because EURO performance depends on many unpredictable factors.

The feature importance charts showed that qualifying goal difference and qualifying goals per match were the most important predictors. Wins per match was the least useful feature, because some teams beat weaker opponents in qualifying but do not repeat that performance in the EURO. The model struggled because qualifying matches involve weaker teams, while the EURO includes only strong teams. There is also a long gap between qualifying and the tournament, during which teams change players, coaches and tactics. The size of the dataset

is also small because the EURO takes place every four years. The high number of missing values also reduces prediction accuracy.

Lessons Learned

Throughout this project, we learned how challenging long-term sports analytics can be when working with historical datasets that contain missing values, inconsistent formats, and limited sample sizes. We gained practical experience in cleaning and merging multiple data sources, standardizing player and team attributes, and building visualizations that highlight trends over time. We also learned that predictive modeling in sports requires careful feature design and that strong performance metrics from qualifying matches do not always translate to tournament success. This project helped us understand the importance of data quality, thoughtful feature engineering, and realistic expectations when modeling unpredictable events.

Future Improvements

Future versions of this project could be improved by incorporating richer and more complete datasets, especially for older tournaments where player attributes and coaching information are sparse. Adding match-level event data such as expected goals (xG), possession, passing accuracy, and shot locations would allow for deeper tactical analysis. A larger dataset that includes other major competitions (World Cup, Nations League, qualifiers across multiple years) could improve model accuracy. More advanced machine learning models, such as gradient boosting or neural networks, could also be tested. Finally, automating the data pipeline would make it easier to update the analysis with new EURO tournaments.

Conclusion

This project examined long-term patterns in EURO tournaments and tested several ideas about player attributes, coaching continuity and predictive modeling. We found a clear rise in goals and attendance over time, while red cards stayed low. Player height and weight patterns matched what is already known about soccer positions, although missing data made the results weaker.

Coaching continuity showed only small effects, and coaching nationality could not be studied because the dataset did not include it. Predicting EURO performance from qualify-

ing performance did not work well because qualifying is very different from tournament play and because the dataset is small.

Even with these challenges, this project helped us understand soccer data more deeply. We learned how to clean long-term historical sports data, how to merge different datasets and how to build visual and statistical analysis around real football trends. The results also showed how difficult it is to predict football outcomes, especially in tournaments where many unpredictable factors play a role.

Team Contributions

Ishmail Khan (khanish2) collected all dataset files, organized directories, and built the initial data loading and cleaning pipeline. He also verified that the merged EURO dataset was accurate and consistent across years.

Mohsin Hossain (hossai34) completed Task 1 by generating visualizations for goals, red cards, and attendance trends. He also performed initial EDA checks to confirm long-term scoring and discipline patterns.

Tawsif Imam Nadif (nadiftaw) handled Task 4 by building the predictive models, evaluating MAE and R^2, and producing feature importance plots. He also integrated all tasks into the final notebook and ensured code reproducibility.

Prachurjo Das (dasprach) worked on Task 2 and Task 3 by grouping player positions, creating height/weight plots, and engineering coaching continuity features. He also assisted with merging lineup and coach data for analysis.

Monirul Amin Mahin (mahinmon) performed data quality checks, created the correlation heatmap, and helped write and format the report. He also reviewed all visualizations and ensured consistent labeling and presentation.

All members contributed to discussions, interpretation of results, and final revisions.

References

UEFA European Championship Dataset (CSV files provided for the project)
Kaggle Discussion Thread (dataset source noted in the problem statement)
Python libraries used: `pandas`, `numpy`, `matplotlib`, `seaborn`, `scikit-learn`.